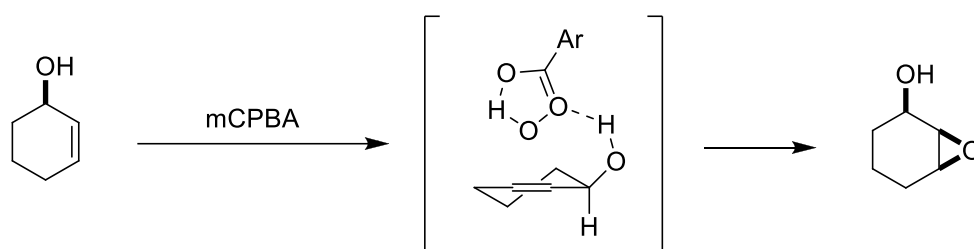


Lecture “Stereochemistry”

Take-home messages 3

7. Directed reactions

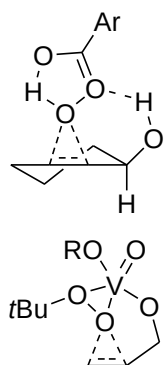
In directed reactions a functional group coordinates or intermediately binds covalently to the reagent and therefore directs the attack to one diastereomeric side of a molecule.



8. Diastereoselective Olefin Transformations

- General aspects:**
- Electrophilic reactions, higher substituted olefins react faster
 - Attack occurs perpendicular to C=C bond, **NOT** along Bürgi-Dunitz trajectory
 - Epoxidations, dihydroxylations and hydroborations are typically stereospecific *syn*-additions

Epoxidations



- Most common reagents (i.e. mCPBA, VO(acac)₂/tBuO₂H) respond nicely to directing effects
- Directing groups include e.g. unprotected alcohols or amides
- Directing groups are essential for epoxidations using VO(acac)₂/tBuO₂H

Dihydroxylations

- OsO₄ is often used in catalytic amounts and Os(VI) reoxidized using NMO etc.
- Dihydroxylations of allylic alcohols (protected or unprotected) are often stereoselective, protecting groups have little influence **EXCEPT** acyl derivatives
- Three models: Vedejs, Houk/Stork, Kishi

Kishi Model: - Find most stable conformer minimizing

A^{1,2}- or A^{1,3}-interactions

- OsO₄ attacks opposite OR group

